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Letters to the Editors

Comments on 'Palaeomagnetism of the Lisbon Volcanics' by N. D. Watkins and A. Richardson

R. Van der Voo

(Received 1968 June 10)

The paper by Watkins & Richardson (1968) provides a detailed study on the paleomagnetic properties of the Lisbon basalt flows. Their Table 5 (page 299) gives an elaborate statistical analysis; the mean direction of magnetization yielded a pole position of 207°E, 69°N (D: 347°, I: +37°, α_{95}: 11°, 39 samples, 12 flows). Particular emphasis was laid on laboratory methods, partly published in another paper as well (Watkins & Haggerty 1967).

It should be noted, however, that Watkins & Richardson's suggestions regarding the undisturbed, flatlying position of the flows are not justified by field evidence. Though the basalt flows are badly exposed, the overlying Oligocene formations can be shown to have a moderate dip towards the east (Lisbon) or southeast (Odivelas, Loures). Examination of the geological maps of the area (Zbyszewski, 1963, 1964) supports this idea. From west to east, on the line Belas–Olivais, we find a successive outcropping of Jurassic, Cretaceous, Eocene basalt flows, Oligocene Benfica formation and Miocene sediments.

The author of the present paper has collected 22 samples from six sites, among which are four sites corresponding with Nos 6, 7, 8 and 9 of Watkins & Richardson, of the same Lisbon volcanics. All samples have been analysed by thermal or a.c. demagnetization in a zero ambient field by methods described previously (As & Zijderveld 1958). A few diagrams are presented here in addition to the decay curves of Watkins & Richardson (Figs 1 and 2). We can observe a similar magnetic behaviour as reported by Watkins & Richardson: Fig. 1 shows two examples of relatively 'hard' N.R.M. (the stable samples of Watkins & Richardson) and Fig. 2 shows a sample with a 'soft' N.R.M. (unstable samples of Watkins & Richardson). Due to the difficulties encountered in the analysis of this soft N.R.M., five samples have been discarded from further analysis. Their directions of magnetization, however, were similar to those of the remaining 17 samples. In Fig. 3 the 'cleaned' directions of magnetization have been plotted after the following corrections have been applied:

correction for the present-day local geomagnetic declination (August 1967: 9·5° W),

correction for the tectonical dip of the strata (up to 15° dip towards the east or southeast).

A mean direction of magnetization has been computed, giving unit weight to samples: D: 354°, I: +40·5° and α_{95}: 4·5°. After unfolding, the precision parameter k increased from 17 to 55. Fig. 4 shows the change in direction of the magnetizations while unfolding. The numerical data have been summarized in Table I. It can easily be seen that no significant difference exists between the present data and the results of Watkins & Richardson, *before* applying the correction for the tectonical dip of the strata.

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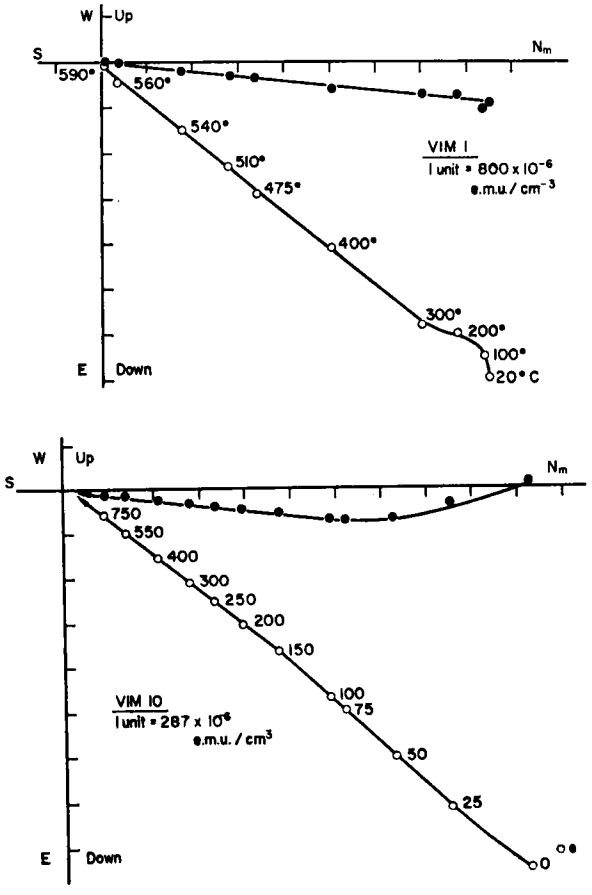


FIG. 1. Diagrams of progressive thermal and a.c. magnetic field demagnetizations of two basalt samples from Lisbon. Plotted points represent successive positions—in orthogonal projection—of the end point of the magnetic vector. Full symbols represent projections on the horizonal plane; open symbols represent projections on the north–south/vertical plane. Not corrected for the present day geomagnetic declination—see text. N_m=magnetic north. Numbers denote temperatures in °C or a.c. field intensities in Oersteds.

Table 1

Summary of the measurements: site means compared with those from Watkins & Richardson

Site	Watkins & Richardson Site No.			Present paper, before T.C.		T.C.	after T.C.	
		D	*I*	*D*	*I*		*D*	*I*
I	—	—	-	5	+41	45/5	10	+43
K	9	354	+46	351	+42	330/5	356	+39
L	—	—	—	351·5	+44·5	0/5	357	+45
M	8	339	+ 8	332	+16	48/19	336	+36·5
N	7	342	+21·5	342	+21	48/15	(348	+33)
O	6	350	+45	339	+31	45/15	345	+44

D and *I* are declination and inclination values.

T.C.: correction for the tectonical dip of the strata.

One result (site N) has been entered in brackets because of the unreliability of the direction of magnetization (see also Fig. 2).

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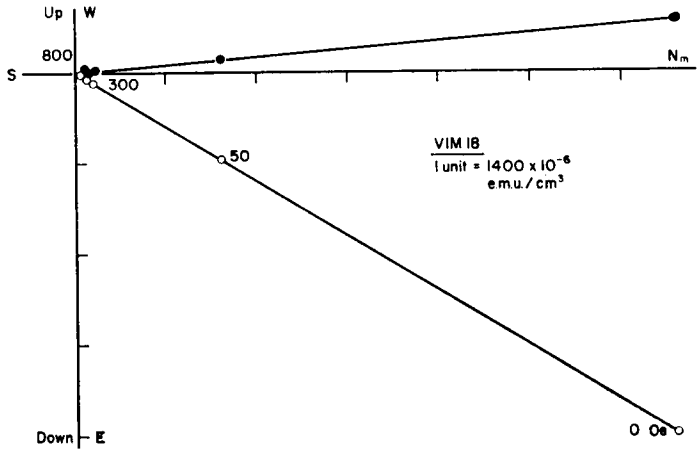


FIG. 2. Demagnetization diagram of a basalt-sample from Odivelas. Further explanation see Fig. 1.

Two other sample groups have been collected from the Sintra intrusive complex (west of Lisbon) and from the Monchique intrusive complex (southern Portugal), radiometrically dated as resp. 80 m.y. and 57 m.y. old (Bonhomme *et al.* 1961; Priem *et al.* 1967). These investigations will be presented as a part of a doctoral thesis (van der Voo, in preparation). I would like to mention them here, however, for comparison with the results obtained from the Lisbon basalts: in Fig. 5 the mean directions of magnetization for the Portuguese rocks are presented as triangles, together with the

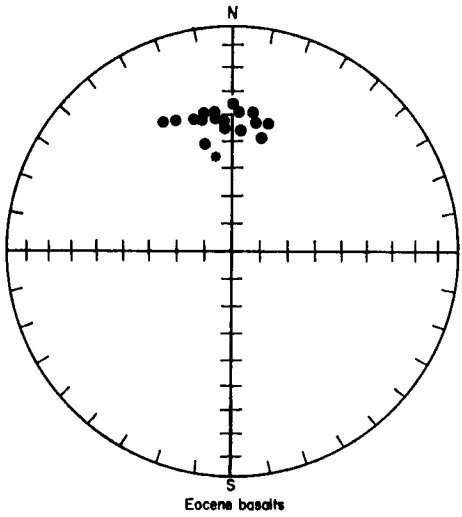


FIG. 3. Equal area projection of the directions of the magnetic vectors, obtained after elimination of the soft (secondary) magnetizations and after tectonic correction. The (full) symbols represent north seeking poles pointing downwards, with a positive inclination. The asterisk denotes the present-day geomagnetic field direction in Portugal.

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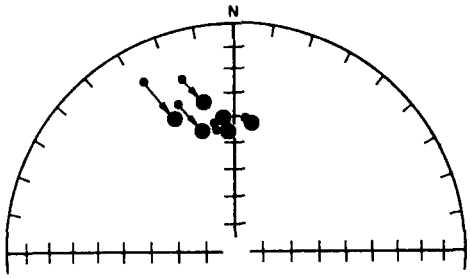


FIG. 4. Equal area projection of the mean directions of magnetization for the sites, obtained from the Lisbon volcanics. Small symbols: before tectonic correction. Large symbols: after tectonic correction.

directions as were to be expected in Portugal, assuming that no relative movements between stable Europe and Portugal have occurred. These directions (dots) have been computed from the virtual pole positions of coeval rocks from stable (extra Alpine) Europe (Irving 1964).

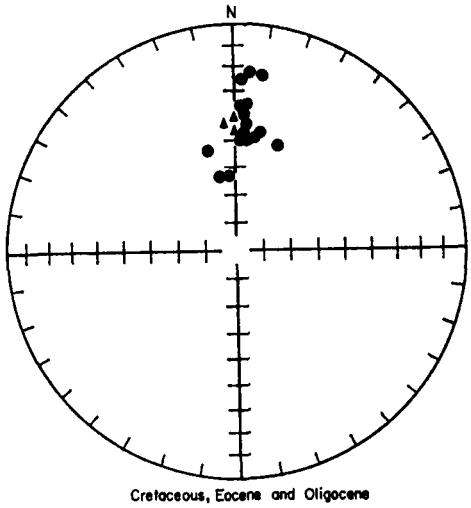


FIG. 5. Equal area projection of the mean directions of magnetization in Portugal obtained from Portuguese rocks (triangles), and rocks from stable Europe (dots). These directions have been computed from the virtual pole positions given by Irving (1964). Further explanation see Fig. 3.

Our conclusion has to be that there is no significant divergence between the Portuguese and the stable European results. The counter-clockwise rotation of the Iberian Meseta, as suggested by palaeomagnetic observations, which has been proven for late Palaeozoic and early Mesozoic times (Girdler 1965; van der Voo 1967; van der Voo in preparation) is likely to have been nearly completed by Eocene times. The *Note added in proof* (Watkins & Richardson 1968, p. 303) on the observations by Mr. E. J. W. Jones, regarding the occurrence of Cretaceous sediment north of Galicia, gives additional evidence for this statement (Jones & Funnel 1968).

Paleomagnetic Laboratory Fort Hoofddijk,
State University of Utrecht,
The Netherlands.

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